Air Force Job Qualification Standard (AFJQS)

Special Experience Identifier (SEI) 3CS

Counter-Small Unmanned Aircraft System (C-sUAS) Operator

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RELEASABILITY: There are no releasability restrictions on this publication.

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1. PURPOSE. This AFJQS identifies tasks and duty requirements for individuals performing duties with the Special Experience Identifier (SEI) 3CS (C-sUAS Operator). It provides the minimum training requirements and standardizes on-the-job training (OJT) tasks necessary to successfully perform duties as a C-sUAS operator. This AFJQS serves as a roadmap to plan and record all C-sUAS duty-position related training, and as a basis for developing a Master Training Plan (MTP) and Individual Training Plan (ITP). This AFJQS must be utilized at all installations with C-sUAS equipment and may be supplemented to capture unique requirements not common across the Total Force. This AFJQS will remain in effect until superseded or rescinded.

2. TASK QUALIFICATION. Personnel will be qualified to the GO/NO GO standard on this AFJQS. GO means the individual can perform the task without assistance utilizing appropriate instructions or other procedural guidance. If required training listed in Part II of this AFJQS is provided through OJT, in-residence, distance learning (DL), or web-based courses, a supervisor/ trainer from that respective functional area and/or a supervisor/trainer assigned as the C-sUAS program manager or C-sUAS Training Noncommissioned Officer in Charge (NCOIC) will complete the trainer's block to document training completion. In order to be awarded the C-sUAS operator SEI 3CS, individuals must be trained and certified on all requirements identified with a single asterisk "*" in the critical task column in Part II of the AFJQS. Explanations for the various Part II Tabs of this AFJQS are as follows:

a. PART II – TAB ONE. AFJQS C-SUAS INITIAL QUALIFICATION TRAINING (IQT). Tasks under this tab are accomplished through completion of a Department of the Air Force (DAF) C-sUAS Formal Training Unit (FTU), or an approved Military Department, vendor, or unit C-sUAS IQT for a specific C-sUAS technology. System-specific C-sUAS IQTs are categorized by the C-sUAS platform type (Command and Control (C2); Fixed-Site; Mobile/Mounted; Containerized/ Portable; Handheld/Dismounted; and Aerial). The tasks in this tab only indicate the result of an individual's successful completion of a C-sUAS IQT for a specific system. For award of the C-sUAS operator SEI 3CS, individuals must successfully

complete a C-sUAS IQT listed in this tab. Not all C-sUAS IQTs are SEI-awarding, refer to the Air Force Enlisted Classification Directory (AFECD) for the approved SEI-awarding C-sUAS IQTs. Completion of a C-sUAS IQT identified in this tab is a prerequisite for the C-sUAS operator tasks in Part II – Tab 2 of this AFJQS.

b. PART II – TAB TWO. AFJQS C-sUAS OPERATOR TRAINING REQUIREMENTS. The tasks in this tab indicate C-sUAS operator tasks to be incorporated into a unit's MTP and C-sUAS mission qualification training (MQT)/C-sUAS Phase II training. These tasks are required for standardization evaluation (Stan Eval) Phase II assessments for certification in the critical duty position of a C-sUAS operator. Sections 1 through 10 of this tab of the AFJQS indicate the core C-sUAS operator tasks required regardless of the C-sUAS IQT completed. Sections 11 through 15 of this tab are the C-sUAS Operations/Battle Drills specific to the applicable C-sUAS platforms and C-sUAS IQT courses listed in Part II – Tab 1 of this AFJQS. The C-sUAS tasks in this tab combined with the tasks identified in Part II – Tab 1 of this AFJQS are required for award of C-sUAS SEI 3CS. Refer to the AFECD for the approved SEI-awarding C-sUAS IQT courses. Completion of a C-sUAS IQT course identified in Part II – Tab 1 is a prerequisite for the C-sUAS operator tasks in this tab.

3. TRAINING DOCUMENTATION AND TRAINER/CERTIFIER IDENTIFICATION. Document training required by this AFJQS in accordance with DAFI 36-2670 and any automated learning management system (LMS) procedural guides. For civilians, document civilian training on supervisor's employee brief (or equivalent) in accordance with AFI 36-129, *Civilian Personnel Management and Administration*. All trainers authorized to sign off tasks in Part II of this AFJQS will be identified in the identification block of this AFJQS. In accordance with DAFI 36-2670, paragraph 4.6.2.2.2., core tasks in this AFJQS are defined as those tasks requiring third-party certification. All tasks identified on this AFJQS require third-party certification for the trainee to be certified. Certifiers will provide third-party certification and evaluation IAW DAFI 36-2670, paragraph 4.6.7.3. If a military or civilian trainee is assigned an additional duty or task not included in an individual's CFETP or this AFJQS, the supervisor will add the task(s) to the MTP and ITP.

4. ADDITIONAL DUTY or TASK. If a C-sUAS operator is assigned an additional duty or task not listed in this AFJQS, the supervisor will develop an AF Form 797, *Job Qualification Standard Continuation/Command JQS* or equivalent. The supervisor, trainer, or C-sUAS program manager will then develop an ITP for the individual to receive training and become duty position certified.

5. RECERTIFYING. For tasks previously certified and required in the current duty position, supervisors/trainers/certifiers will employ the evaluate-train-evaluate process to recertify personnel on tasks. Recertification is not required unless there is a break in the AFJQS task for more than 366 consecutive calendar days. This does not negate the requirement for military and civilian members to undergo annual or otherwise required duty position evaluations, ensuring members are proficient on AFJQS tasks.

6. This AFJQS has been coordinated with all applicable CFMs and MFMs. The overall POC for this AFJQS is AFSFC/S3T AFSFC.S3T.Training@us.af.mil.

BY ORDER OF THE SECRETARY OF THE AIR FORCE

OFFICIAL

DONALD S. GALLAGHER, CMSgt, USAF
Security Forces Career Field Manager

Attachment:
Air Force Job Qualification Standard

This block is for identification purposes only.		
Name of Trainee:		
Printed Name (Last, First, MI)	**Initials (Written)**	**DoD ID (last 4)**
Printed Names and Written Initials of Training and Certifying Officials		
N/I	N/I	
N/I	N/I	
N/I	N/I	
N/I	N/I	
N/I	N/I	
N/I	N/I	
N/I	N/I	

PART II – TAB 1. AFJQS C-sUAS INITIAL QUALIFICATION TRAINING (IQT)						
Critical Tasks	TASKS, KNOWLEDGE AND TECHNICAL REFERENCES	CERTIFICATION				
		START DATE	COMPLETE DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS (IF REQUIRED)
	1. C-SUAS COMMAND AND CONTROL (C2) SYSTEMS TRAINING					
	1.1. Medusa C2 System					
*	1.1.1. Successfully complete Medusa C2 training, if applicable TR(s): AFLCMC/HBUC Medusa C2 operator manual; Applicable technical manuals and commercial publications					
	1.2. Forward Area Air Defense C2 (FAAD C2)					
*	1.2.1. Successfully complete FAAD C2 training, if applicable TR(s): FAAD C2 user manual; Applicable technical manuals and commercial publications					
	1.3. SRC Common Graphical User Interface (GUI)					
*	1.3.1. Successfully complete SRC Common GUI training, if applicable TR(s): SRC Common GUI user manual; Applicable technical manuals and commercial publications					
	1.4. Dowding User Interface (UI)					
*	1.4.1. Successfully complete Dowding UI training, if applicable TR(s): Dowding user manual; Applicable technical manuals and commercial publications					
	1.5. DroneSentry-C2 GUI					
*	1.5.1. Successfully complete DroneSentry-C2 training, if applicable TR(s): DroneShield DroneSentry-C2 GUI user manual; Applicable technical manuals and commercial publications					
	1.6. Windows Tactical Assault Kit (WINTAK)					
*	1.6.1. Successfully complete WINTAK training, if applicable TR(s): WINTAK user manual; Applicable technical manuals and commercial publications					

	1.7. Android Tactical Assault Kit (ATAK)					
*	1.7.1. Successfully complete ATAK training, if applicable TR(s): ATAK user manual; Applicable technical manuals and commercial publications					
	1.8. Other C-sUAS C2 systems/interfaces not previously identified					
*	1.8.1. Successfully complete other C-sUAS C2 systems/interfaces training not previously identified, if applicable TR(s): Applicable technical manuals and commercial publications					
	2. FIXED-SITE C-SUAS IQT COURSES					
	2.1. Medusa System of Systems (SoS)					
*	2.1.1. Successfully complete Medusa operator training, if applicable TR(s): AFLCMC/HBUC Medusa operator manual; Applicable technical manuals and commercial publications					
*	2.1.2. Successfully complete Medusa train-the-trainer (T3) training, if applicable TR(s): AFLCMC/HBUC Medusa operator manual; Applicable technical manuals and commercial publications					
*	2.1.3. Successfully complete Medusa system administrator training, if applicable TR(s): AFLCMC/HBUC Medusa operator manual; Applicable technical manuals and commercial publications					
	2.2. Ninja Gen2 Electronic Attack (EA) system					
*	2.2.1. Successfully complete Ninja basic operator training, if applicable TR(s): Ninja Dashboard User Interface Guide; Applicable technical manuals and commercial publications					
*	2.2.2. Successfully complete Ninja T3 training if applicable TR(s): Ninja Dashboard User Interface Guide; Applicable technical manuals and commercial publications					
*	2.2.3. Successfully complete Ninja advanced training, if applicable TR(s): Ninja Dashboard User Interface Guide; Applicable technical manuals and commercial publications					

	2.3. Fixed-Site - Low-slow-small UAS Integrated Defeat System (FS-LIDS)					
*	2.3.1. Successfully complete FS-LIDS operator training, if applicable TR(s): FS-LIDS user manual; Applicable technical manuals and commercial publications					
*	2.3.2. Successfully complete FAAD C2/FS-LIDS T3 training, if applicable TR(s): FS-LIDS user manual; Applicable technical manuals and commercial publications					
*	2.3.3. Successfully complete FAAD C2/FS-LIDS system administrator training, if applicable TR(s): FS-LIDS user manual; Applicable technical manuals and commercial publications					
	2.4. Joint Counter Radio-Controlled Improvised Explosive Device (RCIED) Electronic Warfare (JCREW) Fixed-Site (AN/FLQ-14(V1))/Drone Restricted Access using Known EW (DRAKE)					
*	2.4.1. Successfully complete JCREW fixed-site (AN/FLQ-14(V)1)/DRAKE operator/administrator/maintainer training, if applicable TR(s): Air Force (AF) Technical Order (T.O.) 31S6-2FLQ14-1, *Technical Manual Operator's Manual for JCREW Fixed-Site Countermeasures Set;* AF T.O. 31S6-2FLQ14-2, *Technical Manual Field Maintenance Repair Part and Special Tools List for JCREW Fixed-Site Countermeasures Set;* and AF T.O. 31S6-3JCEW-1, *JCREW Increment One Block One Mounted, Fixed Site, Dismounted Countermeasures Sets*					
	2.5. DroneSentry					
	2.5.1. Successfully complete DroneSentry operator training, if applicable TR(s): DroneShield DroneSentry user manual; Applicable technical manuals and commercial publications					
	2.5.2. Successfully complete DroneSentry T3 training, if applicable TR(s): DroneShield DroneSentry user manual; Applicable technical manuals and commercial publications					
	2.5.3. Successfully complete DroneSentry system administrator training if applicable					

	TR(s): DroneShield DroneSentry user manual; Applicable technical manuals and commercial publications					
	2.6. Dedrone					
*	2.6.1. Successfully complete Dedrone operator training, if applicable TR(s): Dedrone user manual; Applicable technical manuals and commercial publications					
*	2.6.2. Successfully complete Dedrone T3 training, if applicable TR(s): Dedrone user manual; Applicable technical manuals and commercial publications					
*	2.6.3. Successfully complete Dedrone system administrator training, if applicable TR(s): Dedrone user manual; Applicable technical manuals and commercial publications					
	2.7. Other fixed-site C-sUAS not previously identified					
*	2.7.1. Successfully complete other fixed-site C-sUAS operator, T3, or system administrator training not previously identified, if applicable TR(s): Applicable technical manuals and commercial publications					
	3. MOBILE/MOUNTED C-SUAS IQT COURSES					
	3.1. USAF On-the-Move (OTM) Electromagnetic Warfare (EW) CUAS					
*	3.1.1. Successfully complete USAF OTM EW CUAS operator training, if applicable TR(s): SRC USAF OTM EW CUAS Expanded EW Mission Checklist; Applicable technical manuals and commercial publications					
*	3.1.2. Successfully complete USAF OTM EW CUAS T3 training, if applicable TR(s): SRC USAF OTM EW CUAS Expanded EW Mission Checklist; Applicable technical manuals and commercial publications					
*	3.1.3. Successfully complete USAF OTM EW CUAS system administrator training, if applicable TR(s): SRC USAF OTM EW CUAS Expanded EW Mission Checklist; Applicable technical manuals and commercial publications					

	3.2. Expeditionary Mobile Air Defense Integrated System (X-MADIS)					
*	3.2.1. Successfully complete X-MADIS operator training, if applicable TR(s): Ascent Vision Technologies (AVT) X-MADIS operations manual; Applicable technical manuals and commercial publications					
*	3.2.2. Successfully complete X-MADIS T3 training, if applicable TR(s): AVT X-MADIS operations manual; Applicable technical manuals and commercial publications					
*	3.2.3. Successfully complete X-MADIS system administrator training, if applicable TR(s): AVT X-MADIS operations manual; Applicable technical manuals and commercial publications					
	3.3. High Energy Laser Weapon System (HELWS)					
*	3.3.1. Successfully complete HELWS operator training, if applicable TR(s): Raytheon HELWS Technical Manual D4270694; Raytheon System Operational Training for the HELWS training slides and training support package; Applicable technical manuals and commercial publications					
*	3.3.2. Successfully complete HELWS T3 training, if applicable TR(s): Raytheon HELWS Technical Manual D4270694; Raytheon System Operational Training for the HELWS training slides and training support package; Applicable technical manuals and commercial publications					
*	3.3.3. Successfully complete HELWS system administrator training TR(s): Raytheon HELWS Technical Manual D4270694; Raytheon System Operational Training for the HELWS training slides and training support package; Applicable technical manuals and commercial publications, if applicable					
	3.4. Ninja Mobile					
*	3.4.1. Successfully complete Ninja Mobile operator training, if applicable TR(s): Ninja Dashboard User Interface Guide; Applicable technical manuals and commercial publications					

*	3.4.2. Successfully complete Ninja Mobile T3 training if applicable TR(s): Ninja Dashboard User Interface Guide; Applicable technical manuals and commercial publications					
*	3.4.3. Successfully complete Ninja Mobile advanced training, if applicable TR(s): Ninja Dashboard User Interface Guide; Applicable technical manuals and commercial publications					
	3.5. Mobile LIDS (M-LIDS)					
*	3.5.1. Successfully complete M-LIDS operator training, if applicable TR(s): M-LIDS user manual; Applicable technical manuals and commercial publications					
*	3.5.2. Successfully complete M-LIDS T3 training, if applicable TR(s): M-LIDS user manual; Applicable technical manuals and commercial publications					
*	3.5.3. Successfully complete M-LIDS system administrator training, if applicable TR(s): M-LIDS user manual; Applicable technical manuals and commercial publications					
	3.6. JCREW Mounted AN/ULQ-34(V)1/ DRAKE					
*	3.6.1. Successfully complete JCREW mounted (AN/ULQ-34(V)1)/DRAKE operator/administrator/maintainer training, if applicable TR(s): AF T.O. 31S6-2ULQ34-1, *Technical Manual Field Maintenance Repair Parts and Special Tools List for JCREW Mounted Countermeasures Set;* AF T.O. 31S6-2ULQ34-2, *Technical Manual Operator's Manual for JCREW Mounted Countermeasures Set (AN/ULQ-34(V)1);* and AF T.O. 31S6-3JCEW-1, *JCREW Increment One Block One Mounted, Fixed Site, Dismounted Countermeasures Sets*					
	3.7. Other mobile/mounted C-sUAS not previously identified					
*	3.7.1. Successfully complete other mobile/ mounted C-sUAS operator, T3, or system administrator training not previously identified, if applicable TR(s): Applicable technical manuals and commercial publications					

	4. CONTAINERIZED/PORTABLE C-SUAS IQT COURSES					
	4.1. Containerized Anti-UAS Defense System (C-AUDS)					
*	4.1.1. Successfully complete C-AUDS operator training, if applicable TR(s): Liteye C-AUDS Operator Manual OM-001; Applicable technical manuals and commercial publications					
*	4.1.2. Successfully complete C-AUDS T3 training, if applicable TR(s): Liteye C-AUDS Operator Manual OM-001; Liteye C-AUDS Operator Maintenance Manual 18-1214-OMM; Applicable technical manuals and commercial publications					
*	4.1.3. Successfully complete C-AUDS T3 system administrator training, if applicable TR(s): Liteye C-AUDS Operator Manual OM-001; Liteye C-AUDS Operator Maintenance Manual 18-1214-OMM; Applicable technical manuals and commercial publications					
	4.2. Tactical High Power Microwave Operational Responder (THOR)					
*	4.2.1. Successfully complete THOR operator training, if applicable TR(s): AFRL/SPDE THOR training support package; Applicable technical manuals and commercial publications					
*	4.2.2. Successfully complete THOR T3 training if applicable TR(s): AFRL/SPDE THOR training support package; Applicable technical manuals and commercial publications					
*	4.2.3. Successfully complete THOR system administrator training if applicable TR(s): AFRL/SPDE THOR training support package; Applicable technical manuals and commercial publications					
	4.3. Ninja Agile Combat Employment (ACE) Pallet/SoS					
*	4.3.1. Successfully complete Ninja ACE/ Pallet SoS operator training, if applicable TR(s): AFRL/OWT Global/USAFE-AFAFRICA Ninja ACE training material; Applicable technical manuals and commercial publications					
*	4.3.2. Successfully complete Ninja ACE/ Pallet SoS T3 training, if applicable					

	TR(s): AFRL/OWT Global/USAFE-AFAFRICA Ninja ACE training material; Applicable technical manuals and commercial publications					
*	4.3.3. Successfully complete Ninja ACE/ Pallet SoS system administrator training, if applicable TR(s): AFRL/OWT Global/USAFE-AFAFRICA Ninja ACE training material; Applicable technical manuals and commercial publications					
	4.4. Man-Portable Anti-Drone System-Kit (MADS-K)					
*	4.4.1. Successfully complete MADS-K operator training, if applicable TR(s): Elta MADS-K Operation and Maintenance Manual; MADS-K Kill-Chain Checklist and Best Practices; MADS-K Components Cable Color Coding Checklist; MADS-K Shape File Shape (SHP) Format Map Checklist; Applicable technical manuals and commercial publications					
*	4.4.2. Successfully complete MADS-K T3 training, if applicable TR(s): Elta MADS-K Operation and Maintenance Manual; MADS-K Kill-Chain Checklist and Best Practices; MADS-K Components Cable Color Coding Checklist; MADS-K SHP Format Map Checklist; Applicable technical manuals and commercial publications					
*	4.4.3. Successfully complete MADS-K system administrator training, if applicable TR(s): Elta MADS-K Operation and Maintenance Manual; MADS-K Kill-Chain Checklist and Best Practices; MADS-K Components Cable Color Coding Checklist; MADS-K SHP Format Map Checklist; Applicable technical manuals and commercial publications					
	4.5. Windtalker					
*	4.5.1. Successfully complete Windtalker operator training, if applicable TR(s): Edgesource Windtalker manual; Applicable technical manuals and commercial publications					
*	4.5.2. Successfully complete Windtalker T3 training, if applicable					

	TR(s): Edgesource Windtalker manual; Applicable technical manuals and commercial publications					
*	4.5.3. Successfully complete Windtalker system administrator training, if applicable TR(s): Edgesource Windtalker manual; Applicable technical manuals and commercial publications					
	4.6. Other containerized/portable C-sUAS not previously identified					
*	4.6.1. Successfully complete other containerized/portable C-sUAS operator, T3, or system administrator training not previously identified, if applicable TR(s): Applicable technical manuals and commercial publications					
	5. HANDHELD/DISMOUNTED C-SUAS IQT COURSES					
	5.1. Dronebuster Block 3/Block 3B					
*	5.1.1. Successfully complete Dronebuster Block 3/Block 3B operator training, if applicable TR(s): Radio Hill Technologies Dronebuster Block III User Manual; FlexForce Dronebuster Block 3B operator manual; Applicable technical manuals and commercial publications					
*	5.1.2. Successfully complete Dronebuster Block 3/Block 3B T3 training, if applicable TR(s): Radio Hill Technologies Dronebuster Block III User Manual; FlexForce Dronebuster Block 3B operator manual; Applicable technical manuals and commercial publications					
	5.2. SKYNET Anti-Drone Defense Round					
*	5.2.1. Successfully completed a Remington 12-gauge shotgun/SKYNET Anti-Drone Defense Round training and course of fire, if applicable TR(s): AMTEC Less-Lethal Systems (ALS) Firearms Training Course for the SKYNET Anti-Drone Defense Round manual; Applicable technical manuals and commercial publications					
	5.3. Smart Shooter (SMASH)					
*	5.3.1. Successfully complete SMASH operator training, if applicable TR(s): ELTA SMASH operator and maintenance manual; Applicable technical manuals and commercial publications					

	5.4. Ghoul-Tool Attachable Transmitter (GAT)					
*	5.4.1. Successfully complete GAT operator training, if applicable TR(s): Invisible Interdiction GAT user manual; Applicable technical manuals and commercial publications					
*	5.4.2. Successfully complete GAT T3 training, if applicable TR(s): Invisible Interdiction GAT user manual; Applicable technical manuals and commercial publications					
	5.5. Ninja Dismount					
*	5.5.1. Successfully complete Ninja Dismount operator training, if applicable TR(s): Ninja Dashboard User Interface Guide; Applicable technical manuals and commercial publications					
*	5.5.2. Successfully complete Ninja Dismount T3 training if applicable TR(s): Ninja Dashboard User Interface Guide; Applicable technical manuals and commercial publications					
*	5.5.3. Successfully complete Ninja Dismount advanced training, if applicable TR(s): Ninja Dashboard User Interface Guide; Applicable technical manuals and commercial publications					
	5.6. JCREW Dismounted AN/PLQ-12(V)1/DRAKE					
*	5.6.1. Successfully complete JCREW Dismounted I1B1/DRAKE operator/ administrator/maintainer training, if applicable TR(s): AF T.O. 31S6-2PLQ12-1, *Technical Manual Operator's Manual for JCREW Dismounted Countermeasures Set (AN/PLQ-12(V)1);* AF T.O. 31S6-2PLQ12-11, *Technical Manual Operator's Manual for JCREW Dismounted Countermeasures Set with Dismounted Antenna Mounting System (DAMS) (AN/PLQ-12(V)2);* AF T.O. 31S6-2PLQ12-12, *Technical Manual Field Maintenance Repair Parts and Special Tools List for JCREW Dismounted Countermeasures Set with DAMS (AN/PLQ-12(V)2)*; AF T.O. 31S6-2PLQ12-12-2, *Technical Manual Field Maintenance Repair Parts and Special Tools List for JCREW Dismounted Countermeasures Set (AN/PLQ-*					

	12(V)1); AF T.O. 31S6-2PLQ12-501, *Installation of DAMS on AN/PLQ-12(V)1*; AF T.O. 31S6-2PLQ12-502, *Retrofit Instructions for DAMS-JCREW;* and AF T.O. 31S6-3JCEW-1, *JCREW Increment One Block One Mounted, Fixed Site, Dismounted Countermeasures Sets*					
	5.7. Other handheld/dismounted C-sUAS not previously identified					
*	5.7.1. Successfully complete other handheld/dismounted C-sUAS operator, T3, system administrator training not previously identified, if applicable TR(s): Applicable technical manuals and commercial publications					
	6. AERIAL/LOW COLLATERAL EFFECTS INTERCEPTOR (LCEI) C-SUAS IQT COURSES					
	6.1. Paladin					
*	6.1.1. Successfully complete Paladin operator training, if applicable TR(s): Paladin operator/user manual; AFMAN 11-502, *Small Unmanned Aircraft Systems*; Applicable technical manuals and commercial publications					
*	6.1.2. Successfully complete Paladin T3 training, if applicable TR(s): Paladin operator/user manual; AFMAN 11-502, *Small Unmanned Aircraft Systems*; Applicable technical manuals and commercial publications					
*	6.1.3. Successfully complete Paladin system administrator training, if applicable TR(s): Paladin operator/user manual; AFMAN 11-502, *Small Unmanned Aircraft Systems*; Applicable technical manuals and commercial publications					
	6.2. Drone Hunter					
*	6.2.1. Successfully complete Drone Hunter operator training, if applicable TR(s): Drone Hunter operator/user manual; AFMAN 11-502, *Small Unmanned Aircraft Systems*; Applicable technical manuals and commercial publications					
*	6.2.2. Successfully complete Drone Hunter T3 training, if applicable TR(s): Drone Hunter operator/user manual; AFMAN 11-502, *Small Unmanned Aircraft Systems*; Applicable technical manuals and commercial publications					

*	6.2.3. Successfully complete Drone Hunter system administrator training, if applicable TR(s): Drone Hunter operator/user manual; AFMAN 11-502, *Small Unmanned Aircraft Systems*; Applicable technical manuals and commercial publications					
	6.3. Other aerial/LCEI C-sUAS not previously identified					
*	6.3.1. Successfully complete other aerial/LCEI C-sUAS operator, T3, system administrator training not previously identified, if applicable TR(s): Applicable technical manuals and commercial publications					

PART II – TAB 2. AFJQS C-sUAS OPERATOR TRAINING REQUIREMENTS						
Critical Tasks	TASKS, KNOWLEDGE AND TECHNICAL REFERENCES	CERTIFICATION				
		START DATE	COMPLETE DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS (IF REQUIRED)
	7. SUAS OVERVIEW AND THREATS					
*	7.1. Complete Joint Knowledge Online (JKO) JCSUASA-US1401, Basic sUAS Awareness Training TR(s): https://jkodirect.jten.mil/html/COI.xhtml?course_prefix=JCSUAS&course_number=A-US1401					
*	7.2. Complete JKO JCSUASA-US1402, Basic C-sUAS Systems Familiarization Training TR(s): https://jkodirect.jten.mil/html/COI.xhtml?course_prefix=JCSUAS&course_number=A-US1402					
*	7.3. Complete JKO JCSUAS-US1403, Basic C-sUAS Installation Training TR(s): https://jkodirect.jten.mil/html/COI.xhtml?course_prefix=JCSUAS&course_number=-US1403					
*	7.4. Complete JCSUAS-US1404, Basic C-sUAS TTP Training TR(s): https://jkodirect.jten.mil/html/COI.xhtml?course_prefix=JCSUAS&course_number=-US1404					
*	7.5. Describe the three basic types of sUAS platforms: • Rotary • Fixed wing • Vertical take-off/landing (VTOL) TR(s): AFSFC C-sUAS Updated Planning guidance for SF Commanders; AFTTP 3-4, *Airman's Manual*; DoD C-sUAS Quick Reference Guide (QRG)					
*	7.6. Describe the five UAS group categories and provide the characteristics of the UASs in each group: • Group 1 • Group 2 • Group 3 • Group 4 • Group 5					

	TR(s): AFSFC C-sUAS Updated Planning guidance for SF Commanders; AFTTP 3-4; AFTTP 3-2.64, *Multi-Service Tactics, Techniques, and Procedures (TTPs) for the Tactical Employment of UASs*; AFTTP 3-4.31V1, *Area Security Operations*; DoD C-sUAS QRG; Joint C-sUAS Office (JCO) C-sUAS Playbook; U.S. Army Training Publication (ATP) 3-01.81, *Counter-Unmanned Aircraft System Techniques*					
	7.6.1. Explain which UAS groups the DoD C-sUAS mission is focused on defeating TR(s): AFSFC C-sUAS Updated Planning guidance for SF Commanders; AFTTP 3-4; AFTTP 3-2.64; ATP 3-01.81; DoDD 3800.01E, *DoD Executive Agent for C-sUASs for Unmanned Aircraft Groups 1, 2, and 3*; DoD C-sUAS Strategy					
	7.7. Explain the illicit sUAS activities employed by nefarious actors that pose a threat to US personnel, installations, and missions • Intelligence, surveillance, and reconnaissance (ISR) • Command and control (C2) • Direct attack • Indirect attack • Fire support • Drug smuggling • Prison contraband • Harassment/nuisance TR(s): AFSFC C-sUAS Updated Planning guidance for SF Commanders; AFTTP 3-2.64; DoD C-sUAS QRG; GTA 90-01-047, *Asymmetric Warfare Group (AWG) Unmanned Aerial System (UAS) Threat Awareness Tactical Pocket Reference Guide*; Defense Reduction Agency (DTRA) C-sUAS Smart Card					
	7.8. Explain how sUAS are weaponized: • Internal/affixed improvised explosive devices (IED) • Drop systems (e.g., munitions) • Direct fire systems (e.g., small arms, projectiles, rocket launchers, etc.) • Biological, chemical, radiological TR(s): AFSFC C-sUAS Updated Planning guidance for SF Commanders; AFTTP 3-2.64; DoD C-sUAS QRG					

	7.9. Explain the specific threats/activities that sUAS pose to the local installation or deployment/contingency location TR(s): Local all-hazards threat assessment (AHTA); Antiterrorism vulnerability assessment (ATVA); Integrated Defense Plan (IDP); Intelligence preparation of the battlefield (IPB); Joint intelligence preparation of the operational environment (JIPOE); AFOSI local threat assessment (LTA); Wing Intel					
	7.10. Explain the basic operating frequencies of common sUAS TR(s): Department of Homeland Security (DHS) CUAS-T-G-1, *C-UAS Technology Guide*; DoD C-sUAS QRG; JCO C-sUAS Playbook					
	7.11. Describe the basic components required to operate commercial off-the-shelf (COTS) sUAS TR(s): AFTTP 3-2.64; DHS CUAS-T-G-1; DoD C-sUAS QRG; JCO C-sUAS Playbook;					
	7.12. Explain the four basic features required for fixed-wing and rotary UAS achieve and sustain flight (WEFT): • **W**ings/Rotors/Blades • **E**ngine • **F**uselage • **T**ail TR(s): U.S. Army Training Circular (TC) 3-01.80, *Visual Aircraft Recognition (VACR)*; U.S. Army TC 3-20.31-4, *Direct Fire Engagement Process*					
	7.13. Explain the function of a sUAS radio control (RC) transmitter/ground control station (GCS) TR(s): AFTTP 3-2.64; DHS CUAS-T-G-1; DoD C-sUAS QRG					
	7.14. Explain sUAS telemetry data TR(s): DHS CUAS-T-G-1					
	7.15. Describe manual/direct control sUAS flight mode TR(s): DHS CUAS-T-G-1; DoD C-sUAS QRG					
	7.16. Explain pre-programmed/semi-autonomous sUAS flight mode TR(s): DHS CUAS-T-G-1; DoD C-sUAS QRG					
	7.17. Explain autonomous sUAS flight mode					

	TR(s): DHS CUAS-T-G-1; DoD C-sUAS QRG					
	7.18. Describe the environmental factors that can disrupt sUAS operations: • Wind speed and direction (e.g., headwind, updraft, downdraft) • Air density (e.g., dawn, dusk) • Air particles • Weather (e.g., moisture, rain, fog, low temperatures) • Terrain features • Interference (e.g., radio towers, power lines, large generators, etc.) TR(s): AFTTP 3-2.64					
	7.19. Explain sUAS line of sight (LOS) operations TR(s): AFTTP 3-2.64; DHS CUAS-T-G-1; DoD C-sUAS QRG					
	7.20. Explain sUAS beyond line of sight (BLOS) operations TR(s): AFTTP 3-2.64					
	7.21. Explain sUAS time difference of arrival (TDOA), if applicable TR(s): ACC TRSS EW Fundamentals; DHS CUAS-T-G-1; Applicable technical manuals and commercial publications					
	8. C-SUAS GENERAL KNOWLEDGE					
	8.1. Explain the DoD and AF C-sUAS Mission and Strategy TR(s): AFSFC C-sUAS Updated Planning guidance for SF Commanders; DoDD 3800.01E; DoD C-sUAS Strategy; JCO C-sUAS Playbook					
	8.2. Describe the C-sUAS site selection process TR(s): DoD C-sUAS QRG; DoDD 3222.04, *Electronic Warfare Policy*; CJCSI 3320.02F, *Joint Spectrum Interference Resolution (JSIR)*; CJCSM 3320.02D, *JSIR Procedures*; CJCSM 3212.02E, *Performing Electronic Attack in the US and Canada for Tests, Training and Exercises*; CJCSM 3212.03A, *Performing Tests, Training and Exercises Impacting the Global Positioning System (GPS)*; Joint Publication (JP) 3-85, *Joint Electromagnetic Spectrum Operations (JEMSO)*					
	8.3. Explain a C-sUAS layered defense/ defense in depth architecture					

	TR(s): DoD C-sUAS QRG; JP 3-01, *Countering Air and Missile Threats*; JP 3-85					
	8.4. Explain the gaps in C-sUAS coverage at the local installation or deployment/ contingency location TR(s): Local AHTA; ATVA; IDP; risk assessment					
*	8.5. Explain the C-sUAS kill chain engagement sequence (detect, track, identify, and defeat) TR(s): AFTTP 3-2.64; DHS CUAS-T-G-1; DoD C-sUAS QRG					
*	8.5.1. Describe the general C-sUAS equipment used in the detect engagement sequence of the C-sUAS kill chain (e.g., radar, acoustic sensor, Ninja, Windtalker, WISP, etc.) TR(s): DHS CUAS-T-G-1; DoD C-sUAS QRG; JCO C-sUAS Playbook; Applicable technical manuals and commercial publications					
*	8.5.2. Describe the general C-sUAS equipment used in the track engagement sequence of the C-sUAS kill chain (e.g., radar, acoustic sensor, Ninja, Windtalker, WISP, etc.) TR(s): DHS CUAS-T-G-1; DoD C-sUAS QRG; JCO C-sUAS Playbook; Applicable technical manuals and commercial publications					
*	8.5.3. Describe the general C-sUAS equipment used in the identify sequence of the C-sUAS kill chain (e.g., electro-optical (EO)/infrared (IR) camera TR(s): DHS CUAS-T-G-1; DoD C-sUAS QRG; JCO C-sUAS Playbook; Applicable technical manuals and commercial publications					
*	8.5.4. Describe the general C-sUAS equipment used in the defeat sequence of the C-sUAS kill chain (e.g., EW jammer, Ninja, etc.) TR(s): DHS CUAS-T-G-1; DoD C-sUAS QRG; JCO C-sUAS Playbook; Applicable technical manuals and commercial publications					
	8.6. Explain and provide examples of kinetic C-sUAS countermeasures TR(s): DHS CUAS-T-G-1; DoD C-sUAS QRG; JCO C-sUAS Playbook					

	8.7. Explain and provide examples of non-kinetic C-sUAS countermeasures TR(s): DHS CUAS-T-G-1; DoD C-sUAS QRG; JCO C-sUAS Playbook					
*	8.8. Explain the local C-sUAS engagement zones where sUAS targets will be engaged TR(s): Local/MAJCOM/Combatant Command (CCMD) C-sUAS TTPs; Standard Operating Procedures (SOPs); Checklists; Standing rules of engagement (SROE)/Standing rules for the use of force (SRUF); Special instructions (SPINS); Commander's guidance					
*	8.9. Explain the local procedures on how sUAS targets will be engaged based on the C-sUAS technologies employed TR(s): Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance; C-sUAS-specific operator/user manuals					
*	8.10. Explain the local critical assets to be defended by the C-sUAS in which qualified to operate TR(s): DAFI 31-101, *Integrated Defense (ID)*; Local critical asset list (CAL)/defended asset list (DAL); Integrated Defense Plan (IDP)					
*	8.11. Explain if there are other local C-sUAS technologies on the installation employed and operated by external agencies (e.g., other military services, federal, state, host nation (HN), etc.), if applicable TR(s): Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					
	8.12. Explain and provide examples of active C-sUAS measures TR(s): AFSFC C-sUAS Updated Planning guidance for SF Commanders; AFTTP 3-2.64; ATP 3-01.81; DHS CUAS-T-G-1; DoD C-sUAS QRG					
	8.13. Explain and provide examples of passive C-sUAS measures TR(s): AFSFC C-sUAS Updated Planning guidance for SF Commanders; AFTTP 3-2.64; ATP 3-01.81; DHS CUAS-T-G-1; DoD C-sUAS QRG					
*	8.14. Explain typical sUAS actions during lost link communications disruptions					

	TR(s): AFTTP 3-2.64; DHS CUAS-T-G-1; DoD C-sUAS QRG					
*	8.15. Explain the minimum risk routes (MMRs), including specific pre-planned lost link routes for sUAS recovering autonomously due to loss of their C2 link TR(s): Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					
*	8.16. Explain where the designated safe landing zone is located at the local installation or deployment/contingency location to re-route/land sUAS TR(s): Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					
*	8.17. Explain the local downed sUAS actions, procedures, and notifications TR(s): Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					
*	8.18. Explain the 5Cs for downed sUAS: • **C**onfirm • **C**lear • **C**ordon • **C**ontrol • **C**heck TR(s): AFTTP 3-4; DHS CUAS-T-G-1; DoD C-sUAS QRG; DTRA C-sUAS Smart Card					
*	8.19. Describe CEASE BUZZER/CEASE MUSIC notification requirements and procedures TR(s): CJCSM 3212.02E					
	9. AIR DOMAIN AWARENESS					
*	9.1. Explain airspace control fundamental considerations and basic principles TR(s): AFDP 3-52, *Airspace Control*; AFTTP 3-2.78, *MTTPs for Airspace Control*; AFMAN 11-202, Vol 3, *Flight Operations*; AFMAN 11-214, *Air Operations Rules and Procedures*; AFMAN 11-502, *Small Unmanned Aircraft Systems*; DAFMAN 13-201, *Airspace Management*; JP 3-30, *Joint Air Operations*; JP 3-52, *Joint Airspace Control*					
*	9.2. Explain the National Airspace System (NAS) for CONUS C-sUAS operations					

	TR(s): AFMAN 11-202, AFMAN 11-502; Federal Aviation Administration (FAA) Aeronautical Information Manual (AIM)					
	9.3. Describe the classes of NAS controlled airspaces: • Class A • Class B • Class C • Class D • Class E TR(s): AFMAN 11-202, AFMAN 11-502; FAA AIM					
	9.4. Describe NAS uncontrolled airspace: • Class G TR(s): AFMAN 11-202, AFMAN 11-502; FAA AIM					
	9.5. Describe NAS Special Use Airspace (SUA): • Prohibited areas • Restricted areas • Warning areas • Military operation areas (MOAs) • Alert areas • Controlled firing areas (CFAs) • National security areas (NSAs) TR(s): AFMAN 11-202, AFMAN 11-502; FAA AIM					
	9.6. Describe other NAS airspace areas, for example: • Local airport advisory (LAA) • Military training route (MTR) • Temporary flight restrictions (TFR) • Parachute jump aircraft operations • Air defense identification zones (ADIZ) (land & water based) • Defense VFR (DVFR) flight plan to operate VFR in this airspace • Special security instructions (SSI) TR(s): AFMAN 11-202, AFMAN 11-502; FAA AIM; Title 14 Code of Federal Regulation (CFR) Part 91, *General Operating and Flight Rules*					
*	9.7. Explain FAA notice to air missions (NOTAMs) for the local CONUS installation and the ability to access the information TR(s): FAA AIM; https://notams.aim.faa.gov/notamSearch/nsapp.html#/					

*	9.8. Explain airspace coordination/ deconfliction TR(s): AFDP 3-52; AFTTP 3-2.78; AFMAN 11-202; AFMAN 11-214; DAFMAN 13-201; JP 3-30; JP 3-52					
*	9.9. Explain restricted operating zones (ROZ) TR(s): AFTTP 3-2.31, *M TTPs for Air and Missile Defense*; AFTTP 3-4.31V1; JP 3-52					
*	9.10. Explain airspace control risk considerations TR(s): AFTTP 3-2.31; JP 3-30; JP 3-52					
*	9.11. Explain airspace coordinating measures (ACM) TR(s): AFTTP 3-2.31; AFTTP 3-4.31V1; JP 3-30; JP 3-52					
*	9.12. Explain airspace control authority (ACA) TR(s): AFTTP 3-2.31; AFTTP 3-4.31V1; JP 3-30; JP 3-52					
*	9.13. Explain Title 10 United States Code Section 130i program TR(s): 10 USC § 130i, Protection of certain facilities and assets from unmanned aircraft					
	10. ELECTROMAGNETIC SPECTRUM OPERATIONS (EMSO)					
*	10.1. Explain the fundamentals of the electromagnetic spectrum (EMS) TR(s): ACC Training Support Squadron (TRSS) "Electronic Warfare (EW) Fundamentals"; CJCSM 3212.02E; AF Doctrine Publication (AFDP) 3-51, *Electromagnetic Warfare and Electromagnetic Spectrum Operations*; DAFI 10-706, *Electromagnetic Warfare (EW)*; DAFI 17-220, *Spectrum Management*; DAFI 17-221, *Spectrum Interference Resolution Program*; DoDI 3222.03, *DoD Electromagnetic Environmental Effects (E3) Program*; JP 3-85					
*	10.2. Explain electromagnetic spectrum operations (EMSO) fundamentals TR(s): AFDP 3-51; JP 3-85					
*	10.3. Explain the electromagnetic environment (EME) at the local installation or deployment/contingency location TR(s): AFDP 3-51; DAFI 17-220; JP 3-85; Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SPINS; Applicable technical manuals and commercial publications					

*	10.4. Explain the electromagnetic operational environment (EMOE) at the local installation or deployment/contingency location TR(s): AFDP 3-51; JP 3-85; Local/ MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SPINS; Applicable technical manuals and commercial publications					
*	10.5. Describe the RF spectrum in relation to the EMS TR(s): ACC TRSS EW Fundamentals; CJCSM 3212.03A; CJCSM 3212.02E; DAFI 17-220; DAFI 17-221; JP 3-85					
*	10.6. Describe the fundamentals of radio waves TR(s): ACC TRSS EW Fundamentals; DHS CUAS-T-G-1; JP 3-85					
*	10.7. Explain the common frequencies used by commercial and military UAS (L1 vs L2 band) TR(s): ACC TRSS EW Fundamentals; AFTTP 3-2.64; DHS CUAS-T-G-1; JP 3-85; Applicable technical manuals and commercial publications					
*	10.8. Explain the fundamentals of a modern radar system TR(s): ACC TRSS EW Fundamentals; DHS CUAS-T-G-1; Applicable technical manuals and commercial publications					
	10.8.1. Explain radar characteristics TR(s): ACC TRSS EW Fundamentals; DHS CUAS-T-G-1; Applicable technical manuals and commercial publications					
	10.8.2. Explain radar and pulse-Doppler principles TR(s): ACC TRSS EW Fundamentals; DHS CUAS-T-G-1					
	10.8.3. Explain two-dimensional (2D) radar systems TR(s): DHS CUAS-T-G-1					
	10.8.4. Explain three-dimensional (3D) radar systems TR(s): ACC TRSS EW Fundamentals; DHS CUAS-T-G-1					
*	10.9. Explain the weather impact on C-sUAS sensor operations TR(s): JP 3-85; Applicable technical manuals and commercial publications					
	10.10. Describe antenna characteristics					

	TR(s): ACC TRSS EW Fundamentals; DHS CUAS-T-G-1; Applicable technical manuals and commercial publications					
*	10.11. Describe the effects of EM interference (EMI) TR(s): AFDP 3-51; CJCSI 3320.02F; CJCSM 3320.02D; JP 3-85; DAFI 17-220; DAFI 17-221; DoDI 3222.03; Applicable technical manuals and commercial publications					
	10.12. Explain frequency difference of arrival (FDOA) TR(s): DHS CUAS-T-G-1; Applicable technical manuals and commercial publications					
*	10.13. Explain the three major components of EW TR(s): DAFI 10-706					
*	10.13.1. Explain electromagnetic attack (EA) fundamentals TR(s): ACC TRSS EW Fundamentals; DAFI 10-706; JP 3-85					
*	10.13.2. Explain electromagnetic protection (EP) fundamentals TR(s): ACC TRSS EW Fundamentals; AFDP 3-51; AFI 48-109, *Electromagnetic Field Radiation (EMFR) Occupational and Environmental Health Program*; DAFI 10-706; JP 3-85					
*	10.13.3. Explain EW support (ES) fundamentals TR(s): AFDP 3-51; DAFI 10-706; JP 3-85					
*	10.14. Identify the local C-sUAS engagement authorities for kinetic and non-kinetic EA C-sUAS technologies TR(s): Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance; Applicable technical manuals and commercial publications					
*	10.15. Explain EA/jamming fundamentals TR(s): ACC TRSS EW Fundamentals; DHS CUAS-T-G-1; CJCSM 3212.02E; JP 3-85; Applicable technical manuals and commercial publications					
*	10.16. Explain EW C-sUAS jammer to signal ratio TR(s): ACC TRSS EW Fundamentals; Applicable technical manuals and commercial publications					

*	10.17. Explain local EA request forms for contingency operations, if applicable TR(s): CMCSM 3212.02E; Local/ MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					
*	10.18. Explain local EA request forms for non-contingency operations, if applicable TR(s): CMCSM 3212.02E; Local/ MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					
	11. JOINT COUNTER AIR DEFENSE OPERATIONS					
*	11.1. Explain the C-sUAS mission, roles, responsibilities within the applicable operations center (e.g., Air operations center (AOC), Air defense operations center (ADOC), Base defense operations center (BDOC), Central security control (CSC), Joint operations center (JOC); Tactical operations center (TOC), etc.), where the C-sUAS C2 system is operated TR(s): AFDP 3-01, *Counterair Operations*; AFDP 3-52, *Airspace Control*; AFDP 3-30, *Command and Control*; AFDP 3-60, *Targeting*; AFTTP 3-2.31; AFTTP 3-2.78; AFTTP 3-4.31V3, *Counter Surface to Air Fire*; Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					
*	11.2. Explain integrated air defense system (IADS) TR(s): AFDP 3-01; AFDP 3-60; AFTTP 3-2.31; AFTTP 3-4.31V3; JP 3-01					
*	11.3. Explain the role of the area air defense commander (AADC) TR(s): AFDP 3-01; AFDP 3-30; AFDP 3-52; AFTTP 3-2.31; AFTTP 3-2.78; JP 3-01; JP 3-30; JP 3-52					
*	11.4. Describe an airspace control plan (ACP) TR(s): AFDP 3-01; AFDP 3-52; AFTTP 3-2.31; AFTTP 3-2.78; AFTTP 3-4.31V3; JP 3-01; JP 3-30; JP 3-52					
*	11.5. Describe an air tasking order (ATO) TR(s): AFDP 3-01; AFDP 3-52; AFDP 3-60; AFTTP 3-2.31; AFTTP 3-2.78; JP 3-01; JP 3-30; JP 3-52; JP 3-60					
*	11.6. Describe an airspace control order (ACO)					

	TR(s): AFDP 3-01; AFDP 3-52; AFDP 3-60; AFTTP 3-2.31; AFTTP 3-2.78; JP 3-01; JP 3-30; JP 3-52					
*	11.7. Describe an area air defense plan (AADP) TR(s): AFDP 3-01; AFDP 3-52; AFTTP 3-2.31; AFTTP 3-2.78; AFTTP 3-4.31V3; JP 3-01; JP 3-30; JP 3-52					
*	11.8. Explain weapon engagement zones (WEZ) TR(s): AFTTP 3-2.31; JP 3-01					
*	11.8.1. Explain joint engagement zones (JEZ) TR(s): AFTTP 3-2.31; AFTTP 3-2.78; JP 3-01; JP 3-52					
*	11.8.2. Explain missile engagement zones (MEZ) TR(s): AFTTP 3-2.31; AFTTP 3-2.78; JP 3-01; JP 3-52					
*	11.8.3. Explain fighter engagement zones (FEZ) TR(s): AFTTP 3-2.31; AFTTP 3-2.78; JP 3-01; JP 3-52					
*	11.8.4. Describe the deconfliction/ communication process between fighters, fires (Patriot), and tactical command and control (TAC C2) TR(s): AFTTP 3-2.31; AFTTP 3-2.78; JP 3-01; JP 3-30					
*	11.9. Explain fire support coordination measures (FSCMs) TR(s): AFDP 3-01; AFDP 3-52; AFTTP 3-2.31; AFTTP 3-2.78; AFTTP 3-4.31V1; JP 3-01; JP 3-30; JP 3-52					
*	11.10. Explain the purpose of the weapon control status (WCS) measures for kinetic C-sUAS operations, if applicable TR(s): AFDP 3-52; AFTTP 3-2.31; AFTTP 3-2.78; AFTTP 3-4.31V1; JP 3-01; JP 3-52					
*	11.10.1. Explain the “weapons free” WCS measure TR(s): AFTTP 3-2.31; AFTTP 3-4.31V1; JP 3-01					
*	11.10.2. Explain the “weapons tight” WCS measure TR(s): AFTTP 3-2.31; AFTTP 3-4.31V1; JP 3-01					
*	11.10.3. Explain the “weapons hold” WCS measure					

	TR(s): AFTTP 3-2.31; AFTTP 3-4.31V1; JP 3-01					
*	11.11. Explain the purpose of fire control orders TR(s): AFTTP 3-2.31; JP 3-01					
*	11.11.1. Explain the "engage" fire control order TR(s): AFTTP 3-2.31; JP 3-01					
*	11.11.2. Explain the "cease fire" or "cease engagement" fire control order TR(s): AFTTP 3-2.31; JP 3-01					
*	11.11.3. Explain the "hold fire" fire control order TR(s): AFTTP 3-2.31; JP 3-01					
*	11.12. Explain the theater-specific C-sUAS-related SPINS, if applicable TR(s): Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					
	12. SUAS REPORTING					
*	12.1. Explain the local/MAJCOM/CCMD, and joint staff sUAS initial/operational reporting (OPREP)-3 reporting requirements (as applicable) TR(s): AFSFC C-sUAS Updated Planning guidance for SF Commanders; AFTTP 3-4; Joint Staff sUAS Reporting Execute Order (EXORD) 172100Z Dec 21; Local/ MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					
	12.2. Explain the sUAS reporting acronym DRONE: • **D**etect • **R**eport • **O**bserve • **N**otice • **E**xecute TR(s): AFSFC C-sUAS Updated Planning guidance for SF Commanders; AFTTP 3-4; AFTTP 3-4.31V1					
	12.3. Provide an example of a sUAS SALUTE Report TR(s): AFTTP 3-4; AFTTP 3-4.31V1; DoD C-sUAS QRG					

	13. GROUND OBSERVER VISUAL AIRCRAFT RECOGNITION (VACR)					
	13.1. Describe the physical factors that can influence aircraft detection by a ground observer TR(s): TC 3-01.80, *Visual Aircraft Recognition*					
	13.2. Describe the factors that can affect aircraft identification by a ground observer TR(s): TC 3-01.80					
	13.3. Explain the VACR search techniques ground observers use to help observe and detect sUAS TR(s): TC 3-01.80					
	13.4. Explain the estimation of search limit technique TR(s): TC 3-01.80					
	13.5. Explain vertical scan and search techniques TR(s): ATP 3-01.81; TC 3-01.80; DoD C-sUAS QRG					
	13.6. Explain horizontal scan and search technique TR(s): ATP 3-01.81; TC 3-01.80; DoD C-sUAS QRG					
	13.7. Explain the ground observer siting and offset scan and search techniques TR(s): TC 3-01.80					
	14. C-SUAS GEOSPATIAL REFERENCE SYSTEMS					
	14.1. Demonstrate the ability to use the Military Grid Reference System (MGRS) for the specific C-sUAS in which qualified to operate, if applicable TR(s): CJCSI 3900.01D, *Position (Point and Area) Reference Procedures*; Applicable technical manuals and commercial publications					
	14.2. Demonstrate the ability to use the Global Area Reference System (GARS) for the specific C-sUAS in which qualified to operate, if applicable TR(s): CJCSI 3900.01D; JP 2-03, *Geospatial Intelligence in Joint Operations*; Applicable technical manuals and commercial publications					
	14.3. Demonstrate the ability to determine latitude/longitude (LAT/LONG) for the specific C-sUAS in which qualified to operate, if applicable					

	TR(s): CJCSI 3900.01D; JP 2-03; Applicable technical manuals and commercial publications					
	14.4. Demonstrate the ability to use the Common Geographic Reference System (CGRS) for the specific C-sUAS in which qualified to operate, if applicable TR(s): Applicable technical manuals and commercial publications					
	14.5. Explain the difference between mean sea level (MSL) and above ground level (AGL) TR(s): AFTTP 3-2.78; CJCSI 3900.01D; Applicable technical manuals and commercial publications					
	15. C-SUAS SAFETY AND RISK MANAGEMENT					
*	15.1. Describe the various safety concerns for each C-sUAS in which qualified to operate TR(s): AFI 48-109, *Electromagnetic Field Radiation (EMFR) Occupational and Environmental Health Program*; AFI 91-202, *The US Air Force Mishap Prevention Program*; AFMAN 24-306, *Operation of Air Force Government Motor Vehicles*; DAFMAN 91-203, *Air Force Safety, Fire, and Health Standards*; DAFMAN 91-223, *Aviation Safety Investigations and Reports*; DoDI 3222.03, *DoD Electromagnetic Environmental Effects (E3) Program*; DoDI 6055.01, *DoD Safety and Occupational Health (SOH) Program*; DoDI 6055.05, *Occupational and Environmental Health (OEH)*; DoDI 6055.07, *Mishap Notification, Investigation, Reporting, and Record Keeping*; DoDI 6055.11, *Protecting Personnel from Electromagnetic Fields*; JP 3-85, *Joint Electromagnetic Spectrum Operations*; Applicable technical manuals and commercial publications					
*	15.2. Explain electromagnetic field radiation (EMR) safety for each C-sUAS in which qualified to operate TR(s): AFI 48-109; AFI 91-208, *Hazards of Electromagnetic Radiation to Ordnance (HERO) Certification and Management*; DoDI 3222.03; DoDI 6055.11; JP 3-85; Applicable technical manuals and commercial publications					

*	15.2.1. Explain the hazard of electromagnetic radiation to personnel (HERP) for each C-sUAS in which qualified to operate TR(s): AFI 48-109; DoDI 3222.03; DoDI 6055.11; JP 3-85; Applicable technical manuals and commercial publications					
*	15.2.2. Explain the hazard of electromagnetic radiation to facilities (HERF) for each C-sUAS in which qualified to operate TR(s): AFI 48-109; DoDI 3222.03; JP 3-85; Applicable technical manuals and commercial publications					
*	15.2.3. Explain the hazard of electromagnetic radiation to ordnance (HERO) for each C-sUAS in which qualified to operate TR(s): AFI 48-109; AFI 91-208; DoDI 3222.03; JP 3-85; Applicable technical manuals and commercial publications					
*	15.3. Explain the safe separation distances (SSDs) for each C-sUAS in which qualified to operate TR(s): AFI 91-208; Applicable technical manuals and commercial publications					
*	15.4. Explain directed energy system safety, if applicable TR(s): AFI 91-401, *Directed Energy System Safety*; Applicable technical manuals and commercial publications					
*	15.5. Explain safety hazards associated with downed sUAS (e.g., propellers/propulsion, batteries/power, and payload, etc.) TR(s): DAFMAN 91-223; Applicable technical manuals and commercial publications					
	16. C-SUAS SECURITY REQUIREMENTS					
*	16.1. Demonstrate an understanding of the C-sUAS security classification guide(s) (SCG) for the system in which qualified to operate TR(s): Applicable C-sUAS-specific SCGs; AFRL Mitigation of Small Unmanned Aircraft Systems (MsUAS) SCG; DTRA Counter-Unmanned Systems Air Domain SCG; Directed Energy Weapons (DEWs) SCG (if applicable); DoD Antiterrorism Program and FPCON System SCG;					

	Applicable technical manuals and commercial publications					
*	16.2. Explain the C-sUAS-specific information security (INFOSEC) classification and declassification requirements and procedures TR(s): DoDM 5200.01 Vol 1_AFMAN 16-1404 Vol 1, *Information Security Program: Overview, Classification and Declassification*; Applicable SCGs; technical manuals; and commercial publications					
*	16.3. Explain the marking requirements for the C-sUAS-specific classified information requirements and procedures TR(s): DoDM 5200.01 Vol 2_AFMAN 16-1404 Vol 2, *Information Security Program: Marking of Classified Information*; Applicable SCGs; technical manuals; and commercial publications					
*	16.4. Explain the protection of C-sUAS-specific classified information requirements and procedures TR(s): DoDM 5200.01 Vol 3_AFMAN 16-1404 Vol 3, *Information Security Program: Protection of Classified Information*; Applicable SCGs; technical manuals; and commercial publications					
*	16.5. Explain the C-sUAS-specific operations security (OPSEC) requirements and procedures TR(s): AFI 10-701, *Operations Security*; DoDM 5205.02, *DoD OPSEC Program Manual*; Applicable SCGs; technical manuals; and commercial publications					
*	16.6. Explain the C-sUAS-specific physical security (PHYSEC) requirements and procedures TR(s): DoD 5200.08-R, *Physical Security Program*; DoDM 5200.08 V3_AFMAN31-101 V3, *Installation Perimeter Access Control*; Applicable SCGs; technical manuals; and commercial publications					
*	16.7. Explain the C-sUAS-specific cybersecurity requirements and procedures TR(s): AFI 17-130, *Cybersecurity Program Management*; Applicable SCGs; technical manuals; and commercial publications					
*	16.8. Explain the C-sUAS-specific computer security (COMPUSEC) requirements and procedures					

	TR(s): AFMAN 17-1302-O, *Computer Security*; Applicable SCGs; technical manuals; and commercial publications					
*	16.9. Explain the C-sUAS-specific personnel security (PERSEC) requirements and procedures TR(s): DoDM 5200.02_AFMAN 16-1405, *Air Force Personnel Security Program*; Applicable SCGs; technical manuals; and commercial publications					
*	16.10. Explain the C-sUAS-specific industrial security (INDUSEC) requirements and procedures (if applicable) TR(s): DoDM5220.22 V2_AFMAN16-1406 V2, *National Industrial Security Program: Industrial Security Procedures for Government Activities*; Applicable SCGs; technical manuals; and commercial publications					
*	16.11. Explain the C-sUAS-specific communications security (COMSEC) procedures and requirements TR(s): AFMAN 17-1302-O, *Communications Security Operations*; DoDI 8523.01, *Communications Security*; Applicable SCGs; technical manuals; and commercial publications					
	17. FIXED-SITE C-SUAS OPERATIONS/BATTLE DRILLS *(If employed at the installation or deployment/contingency location)*					
*	17.1. Complete a fixed-site C-sUAS IQT course and maintains currency for the applicable fixed-site C-sUAS listed in Part II – Tab 1 of this AFJQS TR(s): DAFI 36-2646, *Security Forces Training and Standardization Evaluation Programs*; DAFI 36-2670; Applicable fixed-site C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	17.2. Explain the local C-sUAS TTPs, SOPs, SPINS, SROE/SRUF for fixed-site C-sUAS operations in which qualified to operate TR(s): Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					
*	17.3. Demonstrate the ability to initialize the proper startup sequence of the applicable C-sUAS C2 system/interface and operate the applicable fixed-site C-sUAS in which					

	qualified to operate, if C2 capability is integrated as part of the system TR(s): Applicable fixed-site C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	17.4. Identify the main components and explain the nomenclature and characteristics of the applicable fixed-site C-sUAS in which qualified to operate TR(s): Applicable fixed-site C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	17.4.1. Describe the component(s) of the applicable fixed-site C-sUAS used in the detect sequence of the C-sUAS kill chain and how it functions (if component is part of system) TR(s): Applicable fixed-site C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	17.4.2. Describe the component(s) of the applicable fixed-site C-sUAS used in the track sequence of the C-sUAS kill chain and how it functions (if component is part of system) TR(s): Applicable fixed-site C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	17.4.3. Describe the component(s) of the applicable fixed-site C-sUAS used in the identify sequence of the C-sUAS kill chain and how it functions (if component is part of system) TR(s): Applicable fixed-site C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	17.4.4. Describe the component(s) of the applicable fixed-site C-sUAS used in the defeat sequence of the C-sUAS kill chain and how it functions (if component is part of system) TR(s): Applicable fixed-site C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	17.5. Demonstrate the ability to identify the various overlays displayed on the applicable fixed-site C-sUAS C2 system's GUI/UI/ human machine interface (HMI)/ common operating picture (COP)/map display, as applicable:					

	• Protected area • Warning area • Range ring • Inclusion zone • Exclusion zone TR(s): Applicable fixed-site C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	17.6. Demonstrate the ability to operate the various C-sUAS and sub-systems on the applicable fixed-site C-sUAS C2 system's GUI/UI/HMI/COP/map display • C-sUAS (e.g., Ninja, Ninja TRx, etc.) • EW system (e.g., Duke) • Kinetic capability (e.g., Coyote Block 2) • Radar (e.g., AN/TPQ-50, RPS-42, etc.) • EO/IR camera TR(s): Applicable fixed-site C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	17.7. Demonstrate the ability detect a sUAS track on the applicable fixed-site C-sUAS C2 system's GUI/UI/HMI/COP/map display and identify the corresponding system(s) which produced it TR(s): Applicable fixed-site C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	17.8. Demonstrate the ability to track a sUAS on the applicable fixed-site C-sUAS C2 system's GUI/UI/HMI/COP/map display by identifying a line of bearing (LOB) and the corresponding system(s) which produced it TR(s): Applicable fixed-site C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	17.9. Demonstrate the ability to operate and slew the applicable fixed-site C-sUAS EO/IR camera (if camera integrated as part of the system) TR(s): Applicable fixed-site C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	17.10. Demonstrate the ability to make positive identification (PID) utilizing the following: EO/IR camera (if camera					

	integrated as part of the fixed-site system); VACR techniques; or via electronic means by the C-sUAS technology, if applicable TR(s): TC 3-01.80; Applicable fixed-site C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	17.11. Demonstrate the ability to engage and defeat a sUAS with each C-sUAS EW capability integrated in the applicable fixed-site C-sUAS TR(s): Applicable fixed-site C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	17.12. Demonstrate the ability to assess the effectiveness of employed countermeasures (e.g., jamming) utilizing the fixed-site C-sUAS defeat technologies, if applicable TR(s): Applicable fixed-site C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	17.13. Demonstrate the ability to properly shut down the applicable fixed-site C-sUAS C2 system/interface in which qualified to operate TR(s): Applicable fixed-site C-sUAS training material; operator manuals; technical manuals; and commercial publications					
	17.14. Describe the primary and alternate power source(s) for the applicable fixed-site C-sUAS in which qualified to operate TR(s): Applicable fixed-site C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	17.15. Demonstrate the ability to troubleshoot system errors/faults for the applicable fixed-site C-sUAS in which qualified to operate TR(s): Applicable fixed-site C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	17.16. Describe the airspace authorizations and restrictions during the applicable fixed-site C-sUAS operations TR(s): Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					
*	17.17. Explain the EW C-sUAS detection zones during applicable fixed-site C-sUAS operations					

	TR(s): Applicable fixed-site C-sUAS training material; operator manuals; technical manuals; and commercial publications; Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					
*	17.18. Explain EW C-sUAS WEZs during applicable fixed-site C-sUAS operations TR(s): Applicable fixed-site C-sUAS training material; operator manuals; technical manuals; and commercial publications; Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					
*	17.19. Demonstrate the ability to react to degraded communications during the applicable fixed-site C-sUAS operations TR(s): Applicable fixed-site C-sUAS training material; operator manuals; technical manuals; and commercial publications; Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					
	17.20. Explain the role of the MPU5 radio for the applicable fixed-site C-sUAS in which qualified to operate, if applicable TR(s): MPU5 basic operator manual; Applicable fixed-site C-sUAS training material; operator manuals; technical manuals; and commercial publications					
	17.21. Explain the denial and destruction of the applicable fixed-site C-sUAS in which qualified to operate TR(s): Applicable fixed-site C-sUAS training material; operator manuals; technical manuals; and commercial publications; Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					
	17.22. Demonstrate the ability to conduct basic Preventative Maintenance Checks and Services (PMCS) for the applicable fixed-site C-sUAS in which qualified to operate TR(s): Applicable fixed-site C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	17.23. Demonstrate the ability to access, retrieve, and run event logs/reports from the applicable C-sUAS C2 system/interface in					

	which qualified to operate (if the system has the capability) TR(s): Applicable fixed-site C-sUAS training material; operator manuals; technical manuals; and commercial publications					
	18. MOBILE/MOUNTED C-SUAS OPERATIONS/BATTLE DRILLS *(If employed at the installation or deployment/contingency location)*					
*	18.1. Complete a mobile/mounted C-sUAS IQT course and maintains currency for the applicable mobile/mounted C-sUAS listed in Part II – Tab 1 of this AFJQS TR(s): DAFI 36-2646; DAFI 36-2670; Applicable mobile/mounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	18.2. Explain the local C-sUAS TTPs, SOPs, SPINS, and SROE/SRUF for the applicable mobile/mounted C-sUAS operations in which qualified to operate TR(s): Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					
*	18.3. Demonstrate the ability to conduct a pre-mission inspection for the applicable mobile/mounted C-sUAS in which qualified to operate TR(s): AFMAN 24-306; Applicable mobile/mounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	18.4. Demonstrate the ability to initialize the proper startup sequence of the C2 system/ interface and operate the applicable mobile/ mounted C-sUAS in which qualified to operate TR(s): Applicable mobile/mounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	18.5. Identify the main components and explain the nomenclature and characteristics of the applicable mobile/mounted C-sUAS in which qualified to operate TR(s): Applicable mobile/mounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	18.5.1. Describe the component(s) of the applicable mobile/mounted C-sUAS used in the detect sequence of the C-sUAS kill chain and how it functions (if component is part of system)					

	TR(s): Applicable mobile/mounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	18.5.2. Describe the component(s) of the applicable mobile/mounted C-sUAS used in the track sequence of the C-sUAS kill chain and how it functions (if component is part of system) TR(s): Applicable mobile/mounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	18.5.3. Describe the component(s) of the applicable mobile/mounted C-sUAS used in the identify sequence of the C-sUAS kill chain and how it functions (if component is part of system) TR(s): Applicable mobile/mounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	18.5.4. Describe the component(s) of the applicable mobile/mounted C-sUAS used in the defeat sequence of the C-sUAS kill chain and how it functions (if component is part of system) TR(s): Applicable mobile/mounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	18.6. Demonstrate the ability to identify the various overlays displayed on the applicable mobile/mounted C-sUAS C2 system's GUI/ UI/HMI/COP/map display, as applicable TR(s): Applicable mobile/mounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	18.7. Demonstrate the ability to identify the various C-sUAS and sub-systems on the applicable mobile/mounted C-sUAS C2 system's GUI/UI/HMI/COP/map display, as applicable TR(s): Applicable mobile/mounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	18.8. Demonstrate the ability to detect a sUAS track on the applicable mobile/ mounted C-sUAS C2 system's GUI/UI/HMI/ COP/map display and identify the corresponding system(s) which produced it, as applicable					

	TR(s): Applicable mobile/mounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	18.9. Demonstrate the ability to track a sUAS on the applicable mobile/mounted C-sUAS C2 system's GUI/UI/HMI/COP/map display by identifying a LOB and the corresponding system(s) which produced it, as applicable TR(s): Applicable mobile/mounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	18.10. Demonstrate the ability to operate and slew the applicable mobile/mounted C-sUAS EO/IR camera (if camera integrated as part of the system) TR(s): Applicable mobile/mounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	18.11. Demonstrate the ability to make a PID utilizing the following: EO/IR camera (if camera integrated as part of the mobile/ mounted system); VACR techniques; or via electronic means by the C-sUAS technology, if applicable TR(s): TC 3-01.80; Applicable mobile/ mounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	18.12. Demonstrate the ability to engage and defeat a sUAS with each C-sUAS EW capability integrated in the applicable mobile/mounted C-sUAS TR(s): Applicable mobile/mounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	18.13. Demonstrate the ability to assess the effectiveness of employed countermeasures (e.g., jamming) utilizing the applicable mobile/mounted C-sUAS defeat technologies TR(s): Applicable mobile/mounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	18.14. Demonstrate the ability to properly shut down the applicable mobile/mounted C-sUAS in which qualified to operate TR(s): Applicable mobile/mounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					

	18.15. Describe the primary and alternate power source(s) for the applicable mobile/ mounted C-sUAS in which qualified to operate TR(s): Applicable mobile/mounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
	18.16. Demonstrate the ability to troubleshoot system errors/faults for the applicable mobile/mounted C-sUAS in which qualified to operate TR(s): Applicable mobile/mounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	18.17. Demonstrate the ability to maneuver/ direct the tactical movement of the applicable mobile/mounted C-sUAS vehicle and operate the C-sUAS equipment in support of on-the-move C-sUAS operations (e.g., protect resource in transport/convoy operations) TR(s): AFMAN 24-306; Applicable mobile/mounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	18.18. Describe the airspace authorizations and restrictions during the applicable mobile/mounted C-sUAS operations TR(s): Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					
*	18.19. Explain the EW C-sUAS detection zones during applicable mobile/mounted C-sUAS operations TR(s): Applicable mobile/mounted C-sUAS training material; operator manuals; technical manuals; and commercial publications; Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					
*	18.20. Explain EW C-sUAS WEZs during applicable mobile/mounted C-sUAS operations TR(s): Applicable mobile/mounted C-sUAS training material; operator manuals; technical manuals; and commercial publications; Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					

*	18.21. Demonstrate the ability to react to degraded communications during applicable mobile/mounted C-sUAS operations TR(s): Applicable mobile/mounted C-sUAS training material; operator manuals; technical manuals; and commercial publications; Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					
	18.22. Explain the role of the MPU5 radio for the applicable mobile/mounted C-sUAS in which qualified to operate, if applicable TR(s): MPU5 basic operator manual; Applicable mobile/mounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
	18.23. Explain the denial and destruction of the applicable mobile/mounted C-sUAS in which qualified to operate TR(s): Applicable mobile/mounted C-sUAS training material; operator manuals; technical manuals; and commercial publications; Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					
	18.24. Demonstrate the ability to conduct basic PMCS for the applicable mobile/ mounted C-sUAS in which qualified to operate TR(s): Applicable mobile/mounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	18.25. Demonstrate the ability to access, retrieve, and run event logs/reports from the applicable mobile/mounted C2 system/ interface in which qualified to operate (if the system has the capability) TR(s): Applicable mobile/mounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
	19. CONTAINERIZED/PORTABLE C-SUAS OPERATIONS/BATTLE DRILLS *(If employed at the installation or deployment/contingency location)*					
*	19.1. Complete a containerized/portable C-sUAS IQT course and maintains currency for the applicable containerized/portable C-sUAS listed in Part II – Tab 1 of this AFJQS, if applicable TR(s): DAFI 36-2646; DAFI 36-2670; Applicable containerized/portable C-sUAS					

	training material; operator manuals; technical manuals; and commercial publications					
*	19.2. Explain the local C-sUAS TTPs, SOPs, SPINS, SROE/SRUF for the applicable containerized/portable C-sUAS operations in which qualified to operate TR(s): Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					
*	19.3. Demonstrate the ability to initialize the proper startup sequence of the C2 system/ interface and operate the applicable containerized/portable in which qualified to operate TR(s): Applicable containerized/portable C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	19.4. Identify the main components and explain the nomenclature and characteristics of the applicable containerized/portable C-sUAS in which qualified to operate TR(s): Applicable containerized/portable C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	19.4.1. Describe the component(s) of the applicable containerized/portable C-sUAS used in the detect sequence of the C-sUAS kill chain and how it functions (if component is part of system) TR(s): Applicable containerized/portable C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	19.4.2. Describe the component(s) of the applicable containerized/portable C-sUAS used in the track sequence of the C-sUAS kill chain and how it functions (if component is part of system) TR(s): Applicable containerized/portable C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	19.4.3. Describe the component(s) of the applicable containerized/portable C-sUAS used in the identify sequence of the C-sUAS kill chain and how it functions (if component is part of system)					

	TR(s): Applicable containerized/portable C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	19.4.4. Describe the component(s) of the applicable containerized/portable C-sUAS used in the defeat sequence of the C-sUAS kill chain and how it functions (if component is part of system) TR(s): Applicable containerized/portable C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	19.5. Demonstrate the ability to identify the various overlays displayed on the applicable containerized/portable C-sUAS C2 system's GUI/UI/HMI/COP/map display, as applicable TR(s): Applicable containerized/portable C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	19.6. Demonstrate the ability to identify the various C-sUAS and sub-systems on the applicable containerized/portable C-sUAS C2 system's GUI/UI/HMI/COP/map display, as applicable TR(s): Applicable containerized/portable C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	19.7. Demonstrate the ability to detect a sUAS track on the applicable containerized/ portable C-sUAS C2 system's GUI/UI/HMI/COP/map display and identify the corresponding system(s) which produced it, as applicable TR(s): Applicable containerized/portable C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	19.8. Demonstrate the ability to track a sUAS on the containerized/portable C-sUAS C2 system's GUI/UI/HMI/COP/map display by identifying a LOB and the corresponding system(s) which produced it, as applicable TR(s): Applicable containerized/portable C-sUAS training material; operator manuals; technical manuals; and commercial publications					

*	19.9. Demonstrate the ability to operate and slew the applicable containerized/portable C-sUAS EO/IR camera (if camera integrated as part of the system) TR(s): Applicable containerized/portable C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	19.10. Demonstrate the ability to make a PID utilizing the following: EO/IR camera (if camera integrated as part of the containerized/portable system); VACR techniques; or via electronic means by the C-sUAS technology, if applicable TR(s): TC 3-01.80; Applicable containerized/portable C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	19.11. Demonstrate the ability to engage and defeat a sUAS with each C-sUAS EW capability integrated in the applicable containerized/portable C-sUAS TR(s): Applicable containerized/portable C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	19.12. Demonstrate the ability to assess the effectiveness of employed countermeasures (e.g., jamming) utilizing the applicable containerized/portable C-sUAS defeat technologies TR(s): Applicable containerized/portable C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	19.13. Demonstrate the ability to properly shut down the applicable containerized/ portable C-sUAS in which qualified to operate TR(s): Applicable containerized/portable C-sUAS training material; operator manuals; technical manuals; and commercial publications					
	19.14. Describe the primary and alternate power source(s) for the applicable containerized/portable C-sUAS in which qualified to operate TR(s): Applicable containerized/portable C-sUAS training material; operator manuals;					

	technical manuals; and commercial publications					
	19.15. Demonstrate the ability to troubleshoot system errors/faults for the applicable containerized/portable C-sUAS in which qualified to operate TR(s): Applicable containerized/portable C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	19.16. Describe the airspace authorizations and restrictions during the applicable containerized/portable C-sUAS operations TR(s): Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					
*	19.17. Explain the EW C-sUAS detection zones during applicable containerized/ portable C-sUAS operations TR(s): Applicable containerized/portable C-sUAS training material; operator manuals; technical manuals; and commercial publications; Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					
*	19.18. Explain EW C-sUAS WEZs during applicable containerized/portable C-sUAS operations TR(s): Applicable containerized/portable C-sUAS training material; operator manuals; technical manuals; and commercial publications; Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					
*	19.19. Demonstrate the ability to react to degraded communications during applicable containerized/portable C-sUAS operations TR(s): Applicable containerized/portable C-sUAS training material; operator manuals; technical manuals; and commercial publications; Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/ SRUF; SPINS; Commander's guidance					
	19.20. Explain the role of the MPU5 radio for the applicable containerized/portable C-sUAS in which qualified to operate, if applicable					

	TR(s): MPU5 basic operator manual; Applicable containerized/portable C-sUAS training material; operator manuals; technical manuals; and commercial publications					
	19.21. Explain the denial and destruction of the applicable containerized/portable C-sUAS in which qualified to operate TR(s): Applicable containerized/portable C-sUAS training material; operator manuals; technical manuals; and commercial publications; Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/ SRUF; SPINS; Commander's guidance					
	19.22. Demonstrate the ability to conduct basic PMCS for the applicable containerized/ portable C-sUAS in which qualified to operate TR(s): Applicable containerized/portable C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	19.23. Demonstrate the ability to access, retrieve, and run event logs/reports from the applicable containerized/portable C2 system in which qualified to operate (if the system has the capability) TR(s): Applicable containerized/portable C-sUAS training material; operator manuals; technical manuals; and commercial publications					
	20. HANDHELD/DISMOUNTED C-SUAS OPERATIONS/BATTLE DRILLS ***(If employed at the installation or deployment/contingency location)***					
*	20.1. Complete a handheld/dismounted C-sUAS IQT course and maintains currency for the applicable handheld/dismounted C-sUAS listed in Part II – Tab 1 of this AFJQS TR(s): DAFI 36-2646; DAFI 36-2670; Applicable handheld/dismounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	20.2. Explain the local C-sUAS TTPs, SOPs, SPINS, SROE/SRUF for the handheld/dismounted C-sUAS operations in which qualified to operate TR(s): Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					
*	20.3. Demonstrate the ability to operate the applicable handheld/dismounted C-sUAS in which qualified to operate					

	TR(s): Applicable handheld/dismounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	20.4. Explain the C-sUAS kill chain engagement functions and procedures of the applicable handheld/dismounted C-sUAS in which qualified to operate TR(s): Applicable handheld/dismounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	20.5. Identify the main components and explain the nomenclature and characteristics of the applicable handheld/dismounted C-sUAS in which qualified to operate TR(s): Applicable handheld/dismounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	20.5.1. Describe the component of the applicable handheld/dismounted C-sUAS used in the detect sequence of the C-sUAS kill chain and how it functions (if component is part of system) TR(s): Applicable handheld/dismounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	20.5.2. Describe the component of the applicable handheld/dismounted C-sUAS used in the track sequence of the C-sUAS kill chain and how it functions (if component is part of system) TR(s): Applicable handheld/dismounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	20.5.3. Describe the component of the applicable handheld/dismounted C-sUAS used in the identify sequence of the C-sUAS kill chain and how it functions (if component is part of system) TR(s): Applicable handheld/dismounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	20.5.4. Describe the component of the applicable handheld/dismounted C-sUAS used in the defeat sequence of the C-sUAS					

	kill chain and how it functions (if component is part of system) TR(s): Applicable handheld/dismounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	20.6. Explain the jamming capabilities (e.g., C2, GPS) of the applicable handheld/ dismounted C-sUAS in which qualified to operate TR(s): Applicable handheld/dismounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	20.7. Describe the primary and alternate power source(s) for the applicable handheld/ dismounted C-sUAS in which qualified to operate TR(s): Applicable handheld/dismounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	20.8. Demonstrate the ability to troubleshoot system errors/faults for the applicable handheld/dismounted C-sUAS in which qualified to operate TR(s): Applicable handheld/dismounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	20.9. Demonstrate the ability to aim and engage a sUAS utilizing the applicable handheld/dismounted C-sUAS in which qualified to operate TR(s): Applicable handheld/dismounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	20.10. Demonstrate the ability to assess the effectiveness of employed countermeasures (e.g., jamming) utilizing the applicable handheld/dismounted C-sUAS defeat technologies TR(s): Applicable handheld/dismounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	20.11. Explain the maximum effective range of the applicable handheld/dismounted C-sUAS in which qualified to operate					

	TR(s): Applicable handheld/dismounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
*	20.12. Describe the airspace authorizations and restrictions during the applicable handheld/dismounted C-sUAS operations TR(s): Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					
*	20.13. Explain the C-sUAS detection zones during the applicable handheld/dismounted C-sUAS operations TR(s): Applicable handheld/dismounted C-sUAS training material; operator manuals; technical manuals; and commercial publications; Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/ SRUF; SPINS; Commander's guidance					
	20.14. Explain C-sUAS WEZs during the applicable handheld/dismounted C-sUAS operations TR(s): Applicable handheld/dismounted C-sUAS training material; operator manuals; technical manuals; and commercial publications; Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/ SRUF; SPINS; Commander's guidance					
*	20.15. Demonstrate the ability to react to degraded communications during the handheld/dismounted C-sUAS operations TR(s): Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					
	20.16. Explain the role of the MPU5 radio for the applicable handheld/dismounted C-sUAS in which qualified to operate, if applicable TR(s): MPU5 basic operator manual; Applicable handheld/dismounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
	20.17. Explain the denial and destruction of the applicable handheld/dismounted C-sUAS in which qualified to operate TR(s): Applicable handheld/dismounted C-sUAS training material; Applicable C-sUAS operator manuals; technical manuals; and commercial publications; Local/MAJCOM/ CCMD C-sUAS TTPs; SOPs; Checklists;					

	SROE/SRUF; SPINS; Commander's guidance					
	20.18. Demonstrate the ability to conduct basic PMCS for the applicable handheld/ dismounted C-sUAS in which qualified to operate TR(s): Applicable handheld/dismounted C-sUAS training material; operator manuals; technical manuals; and commercial publications					
	21. AERIAL C-SUAS OPERATIONS/BATTLE DRILLS *(If employed at the installation or deployment/contingency location)*					
*	21.1. Complete an aerial C-sUAS IQT course and maintains currency for the applicable aerial C-sUAS listed in Part II – Tab 1 of this AFJQS TR(s): AFMAN 11-502; DAFI 36-2646; DAFI 36-2670; Applicable aerial C-sUAS training material; sUAS/C-sUAS operator manuals; flight manuals; technical manuals; and commercial publications					
*	21.2. Meets all mandatory AF sUAS operator training and certification requirements TR(s): AFMAN 11-502; Title 14 CFR Part 107, *Small Unmanned Aircraft Systems* (as applicable)					
*	21.3. Explain the local C-sUAS TTPs, SOPs, SPINS, SROE/SRUF for the applicable aerial C-sUAS operations (e.g., Paladin) in which qualified to operate TR(s): Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					
*	21.4. Demonstrate the ability to conduct live flight launch and recovery operations for the applicable aerial C-sUAS in which qualified to operate TR(s): AFMAN 11-502; Applicable aerial C-sUAS training material; sUAS/C-sUAS operator manuals; flight manuals; technical manuals; and commercial publications					
*	21.5. Explain the C-sUAS kill chain engagement functions and procedures of the applicable aerial C-sUAS in which qualified to operate TR(s): Applicable aerial C-sUAS training material; sUAS/C-sUAS operator manuals; flight manuals; technical manuals; and commercial publications					

*	21.6. Identify the main components and explain the nomenclature and characteristics of the applicable aerial C-sUAS in which qualified to operate TR(s): Applicable aerial C-sUAS training material; sUAS/C-sUAS operator manuals; flight manuals; technical manuals; and commercial publications					
*	21.6.1. Describe the component of the applicable aerial C-sUAS used in the detect sequence of the C-sUAS kill chain and how it functions (if component is part of system) TR(s): Applicable aerial C-sUAS training material; sUAS/C-sUAS operator manuals; flight manuals; technical manuals; and commercial publications					
*	21.6.2. Describe the component of the applicable aerial C-sUAS used in the track sequence of the C-sUAS kill chain and how it functions (if component is part of system) TR(s): Applicable aerial C-sUAS training material; sUAS/C-sUAS operator manuals; flight manuals; technical manuals; and commercial publications					
*	21.6.3. Describe the component of the applicable aerial C-sUAS used in the identify sequence of the C-sUAS kill chain and how it functions (if component is part of system) TR(s): Applicable aerial C-sUAS training material; sUAS/C-sUAS operator manuals; flight manuals; technical manuals; and commercial publications					
*	21.6.4. Describe the component of the applicable aerial C-sUAS used in the defeat sequence of the C-sUAS kill chain and how it functions (if component is part of system) TR(s): Applicable aerial C-sUAS training material; sUAS/C-sUAS operator manuals; flight manuals; technical manuals; and commercial publications					
	21.7. Describe the primary and alternate power source(s) for the applicable aerial C-sUAS in which qualified to operate TR(s): Applicable aerial C-sUAS training material; sUAS/C-sUAS operator manuals; flight manuals; technical manuals; and commercial publications					
*	21.8. Demonstrate the ability to engage and defeat a sUAS utilizing the applicable aerial C-sUAS in which qualified to operate					

	TR(s): Applicable aerial C-sUAS training material; sUAS/C-sUAS operator manuals; flight manuals; technical manuals; and commercial publications					
*	21.9. Demonstrate the ability to assess the effectiveness of employed countermeasures (e.g., net capture) utilizing the applicable aerial C-sUAS in which qualified to operate, if applicable TR(s): Applicable aerial C-sUAS training material; sUAS/C-sUAS operator manuals; flight manuals; technical manuals; and commercial publications					
	21.10. Demonstrate the ability to troubleshoot system errors/faults for the applicable aerial C-sUAS in which qualified to operate TR(s): Applicable aerial C-sUAS training material; sUAS/C-sUAS operator manuals; flight manuals; technical manuals; and commercial publications					
*	21.11. Explain the maximum effective range and flight time of the applicable aerial C-sUAS in which qualified to operate TR(s): Applicable aerial C-sUAS training material; sUAS/C-sUAS operator manuals; flight manuals; technical manuals; and commercial publications					
*	21.12. Describe the airspace authorizations and restrictions during for the applicable aerial C-sUAS operations TR(s): Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					
*	21.13. Explain the C-sUAS detection zones during applicable aerial C-sUAS operations TR(s): Applicable aerial C-sUAS training material; sUAS/C-sUAS operator manuals; flight manuals; technical manuals; and commercial publications; Local/MAJCOM/ CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					
*	21.14. Explain aerial C-sUAS WEZs during the applicable aerial C-sUAS operations TR(s): Applicable aerial C-sUAS training material; sUAS/C-sUAS operator manuals; flight manuals; technical manuals; and commercial publications; Local/MAJCOM/ CCMD C-sUAS TTPs; SOPs; Checklists;					

	SROE/SRUF; SPINS; Commander's guidance					
*	21.15. Demonstrate the ability to react to degraded communications during the applicable aerial C-sUAS operations TR(s): Applicable aerial C-sUAS training material; sUAS/C-sUAS operator manuals; flight manuals; technical manuals; and commercial publications; Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					
	21.16. Explain the role of the MPU5 radio for the applicable aerial C-sUAS in which qualified to operate, if applicable TR(s): MPU5 basic operator manual; Applicable aerial C-sUAS training material; sUAS/C-sUAS operator manuals; flight manuals; technical manuals; and commercial publications					
	21.17. Explain the denial and destruction of aerial C-sUAS in which qualified to operate TR(s): Applicable aerial C-sUAS training material; sUAS/C-sUAS operator manuals; flight manuals; technical manuals; and commercial publications; Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					
	21.18. Demonstrate the ability to conduct basic PMCS for the applicable aerial C-sUAS in which qualified to operate TR(s): Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					
*	21.19. Explain the aviation safety requirements for the applicable aerial C-sUAS in which qualified to operate TR(s): AFMAN 11-502; DAFMAN 91-223; Local/MAJCOM/CCMD C-sUAS TTPs; SOPs; Checklists; SROE/SRUF; SPINS; Commander's guidance					